

The empirical recurrence for OEIS A302474, recovered from its tail

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Abstract

OEIS A302474 records “Empirical recurrence of order 63” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 67$. Its last coefficient is $c_{63} = 48 \neq 0$, so no further tail satisfies anything shorter; any order-63 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302474 is “Number of $4 \times n$ 0..1 arrays with every element equal to 1, 2, 3 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 34, 250, 1401, 10264, 78679, 584333, 4330427, ...

The entry states:

Empirical recurrence of order 63 (see link above).

As of the “Last modified” line on the live entry (Aug 02 2025 12:16:28, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 63: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 63.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 4$ on, it returns order 63 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	2	c_{17}	−37744688	c_{33}	2800686077	c_{49}	23474928
c_2	53	c_{18}	34891410	c_{34}	−1957542288	c_{50}	50220931
c_3	29	c_{19}	33333973	c_{35}	3802056498	c_{51}	−29795142
c_4	−780	c_{20}	−26019671	c_{36}	−4666822440	c_{52}	10433854
c_5	−1539	c_{21}	259675717	c_{37}	−4946039433	c_{53}	−12641507
c_6	1959	c_{22}	−112956284	c_{38}	777313434	c_{54}	2826971
c_7	10472	c_{23}	−442582452	c_{39}	−1506347806	c_{55}	1709830
c_8	37548	c_{24}	419822922	c_{40}	3091652009	c_{56}	−66020
c_9	51586	c_{25}	−920585585	c_{41}	1757922341	c_{57}	−232899
c_{10}	−298324	c_{26}	−630702931	c_{42}	−105514378	c_{58}	85314
c_{11}	−739793	c_{27}	1750328884	c_{43}	419020297	c_{59}	−9480
c_{12}	798578	c_{28}	−1481516871	c_{44}	−732896908	c_{60}	−3010
c_{13}	1915840	c_{29}	1033082140	c_{45}	−95234748	c_{61}	300
c_{14}	−282424	c_{30}	2685611286	c_{46}	−155656407	c_{62}	−132
c_{15}	5155143	c_{31}	−3945079477	c_{47}	−20765812	c_{63}	48
c_{16}	−7841867	c_{32}	2706735130	c_{48}	14096073		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 67$, and it is the recurrence OEIS A302474 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{63} - c_1 x^{63-1} - \dots - c_{63}$. Its constant term is $-c_{63} = -48$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 63 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 63, so $\deg q = 63$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 63, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 63, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 48.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-63 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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